

SpotOn

Progress #4



Agenda

- **Recap**
- **Tracker / Re-ID model comparison**
- **Similarity Search**
- **Workflow refinement**
- **Interesting topic from the final exam AI-Enabled**
- **Frontend / Backend communication**
- **Internship plan**

Recap

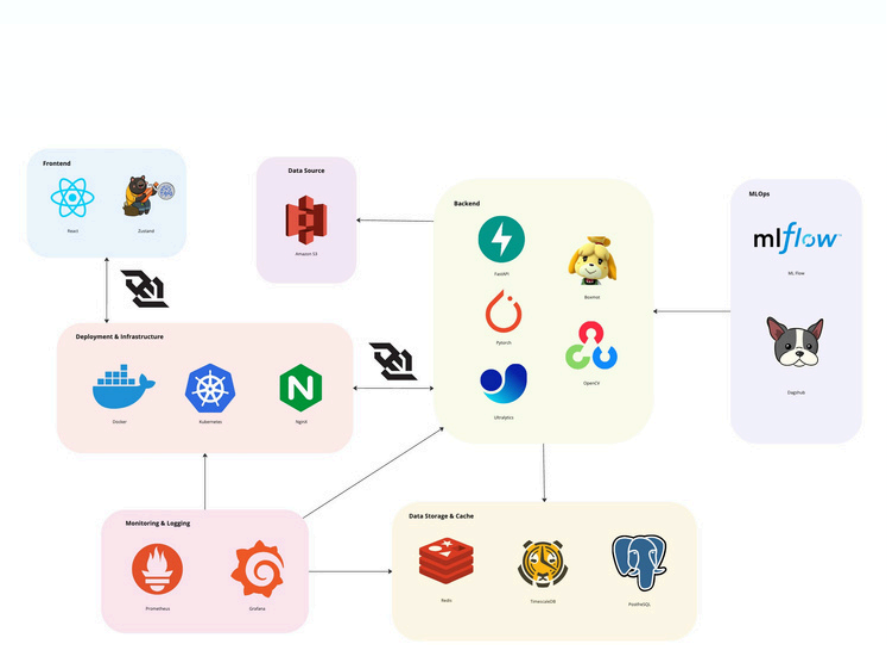
ML Deployment

Updated Architecture

Tracking Logic

Deployment Comparing

Embedded Model (in Backend)	Separate ML Server (MLflow via HTTPS)
~0 ms (only inference time matters)	~100 ms - 300 ms+ (just for communication)
Limited primarily by slowest inference step	Severely limited by communication overhead
Budget: 200ms/frame. If inference < 200ms, OK.	Budget: 200ms/frame. Communication alone (100-300ms+) likely exceeds budget, leaving no time for inference.
Budget: 500ms/frame. Inference likely fits. OK.	Budget: 500ms/frame. Communication (100-300ms+) consumes 20-60%+ of budget. Might be possible, but not ideal.
Backend+Model scale together	Independent ML scaling (Good)
Tighter coupling	Better separation (Good)
Backend needs CPU + GPU	Backend (CPU), ML Server (CPU+GPU)



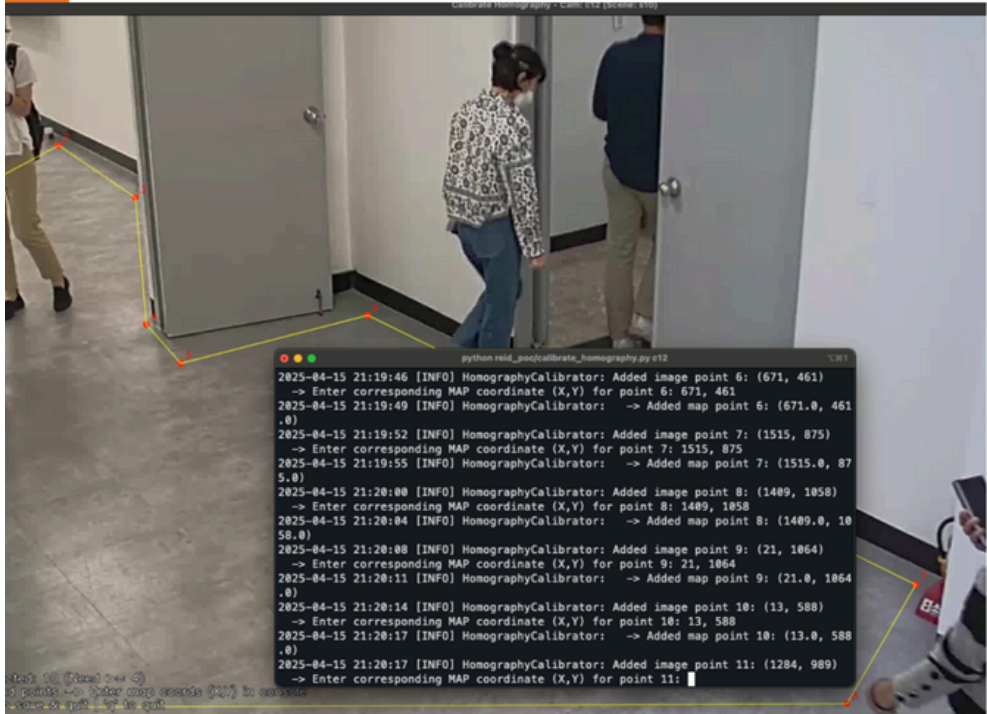
Solution #1

Strategies

- Modify the ByteTrack
- Batch Processing
- Use Half Precision
- Reduce Input Resolution
- Compile the model into faster format (ONNX/TensorRT)
- Detection frame skipping
- Switch to faster detection model

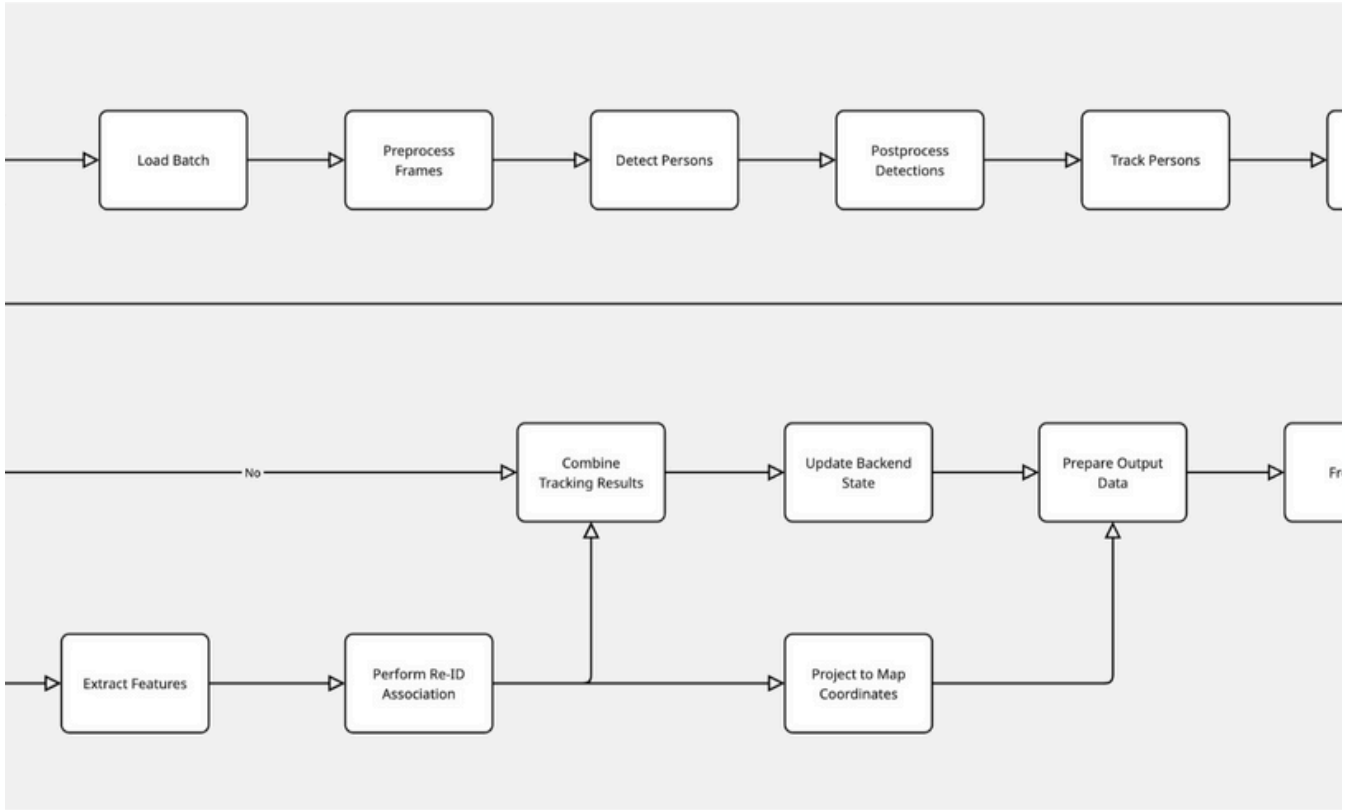
Effective Untested Unsuccessful

Unified Spatial Mapping



Calibrate
Homograph
(Getting the ground

Data Flow



Tracker / Re-ID model comparison

Trackers

Tracker	Status	HOTA↑	MOTA↑	IDF1↑	FPS
boosttrack	✓	69.253	75.914	83.206	25
botsort	✓	68.885	78.222	81.344	46
strongsort	✓	68.05	76.185	80.763	17
bytetrack	✓	67.68	78.039	79.157	1265
deepocsort	✓	67.509	75.83	79.976	12
ocsort	✓	66.441	74.548	77.899	1483

ReID models

- clip_duke.pt
- clip_market1501.pt
- lmbn_n_duke.pt
- osnet_ibn_x1_0_msmt17.pt
- osnet_ain_x1_0_msmt17.pt
- osnet_x1_0_msmt17.pt
- mobilenetv2_x1_4_market1501.pt
- mobilenetv2_x1_4_dukemtmcrid.pt

Experiments

SpotOn

Tracking+ReID Comparison (BoxMOT) ⓘ Provide Feedback ↗ Add Description Share

Runs Evaluation Experimental Traces

⌵

🔍

metrics.rmse < 1 and params.model = "tree" ⓘ

Time created ▾

State: Active ▾

Datasets ▾

⋮

+ New run

⌵ Sort: Created ▾

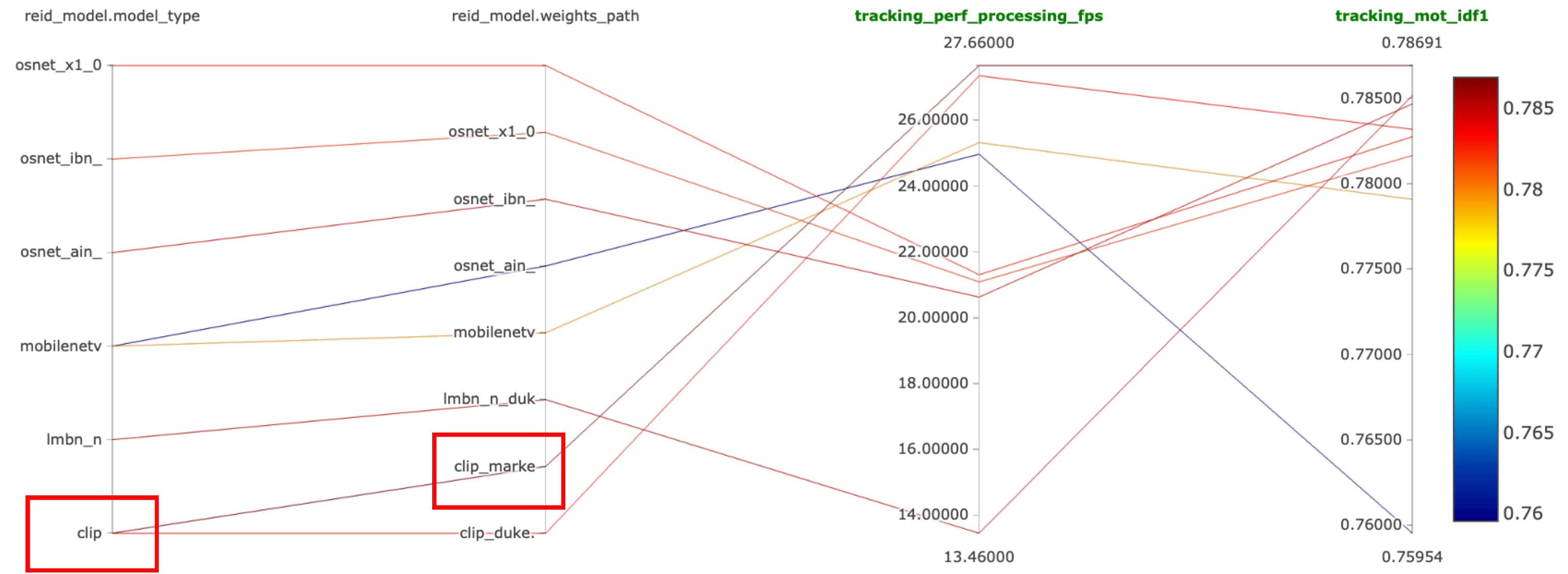
📄 Columns ▾

📄 Group by ▾

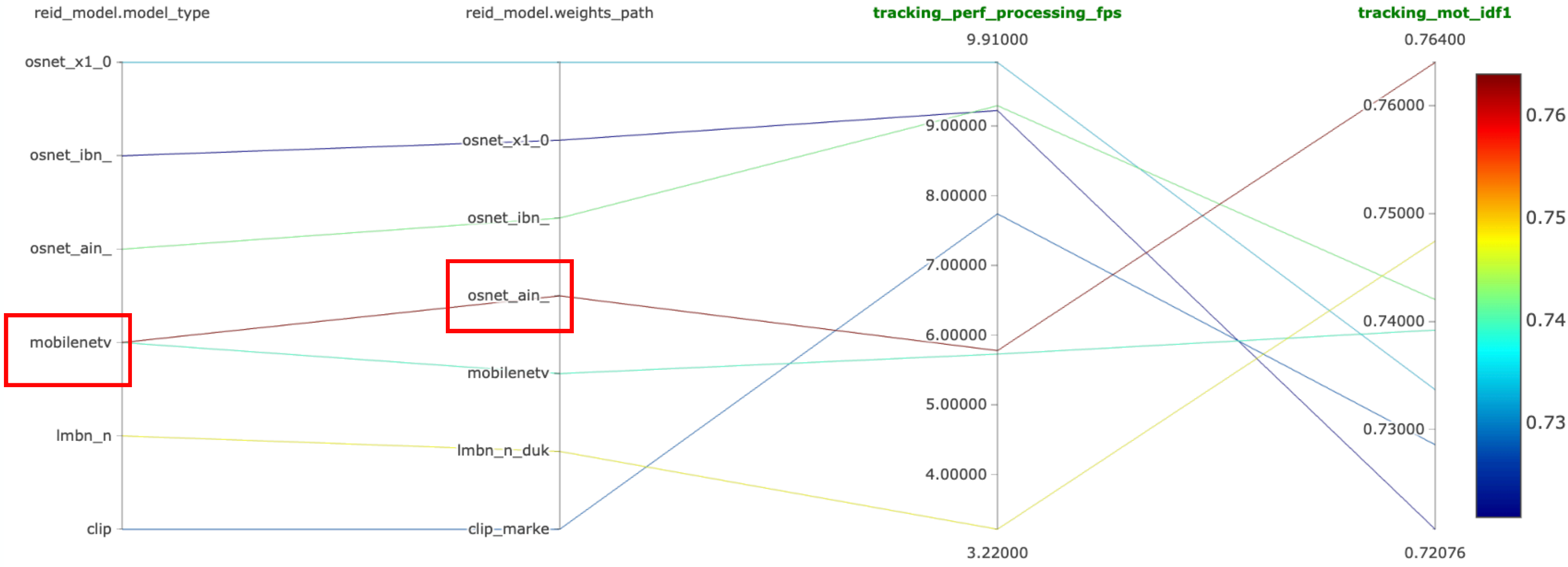
☐	Run Name	Created ⌵	Dataset
☐	☐ 📌 ☐ ● mtmmc_tracking_reid_comparison	✓ 25 days ago	-
☐	● tracker_imprassoc_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_dukemtcreid	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_market1501	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_osnet_x1_0_osnet_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_osnet_ain_x1_0_osnet_ain_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_osnet_ibn_x1_0_osnet_ibn_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_lmbn_n_lmbn_n_duke	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_clip_clip_market1501	✓ 24 days ago	-
☐	● tracker_imprassoc_reid_clip_clip_duke	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_dukemtcreid	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_market1501	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_osnet_x1_0_osnet_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_osnet_ain_x1_0_osnet_ain_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_osnet_ibn_x1_0_osnet_ibn_x1_0_msmt17	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_lmbn_n_lmbn_n_duke	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_clip_clip_market1501	✓ 24 days ago	-
☐	● tracker_boosttrack_reid_clip_clip_duke	✓ 24 days ago	-
☐	● tracker_botsort_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_dukemtcreid	✓ 24 days ago	-
☐	● tracker_botsort_reid_mobilenetv2_x1_4_mobilenetv2_x1_4_market1501	✓ 24 days ago	-

101 matching runs

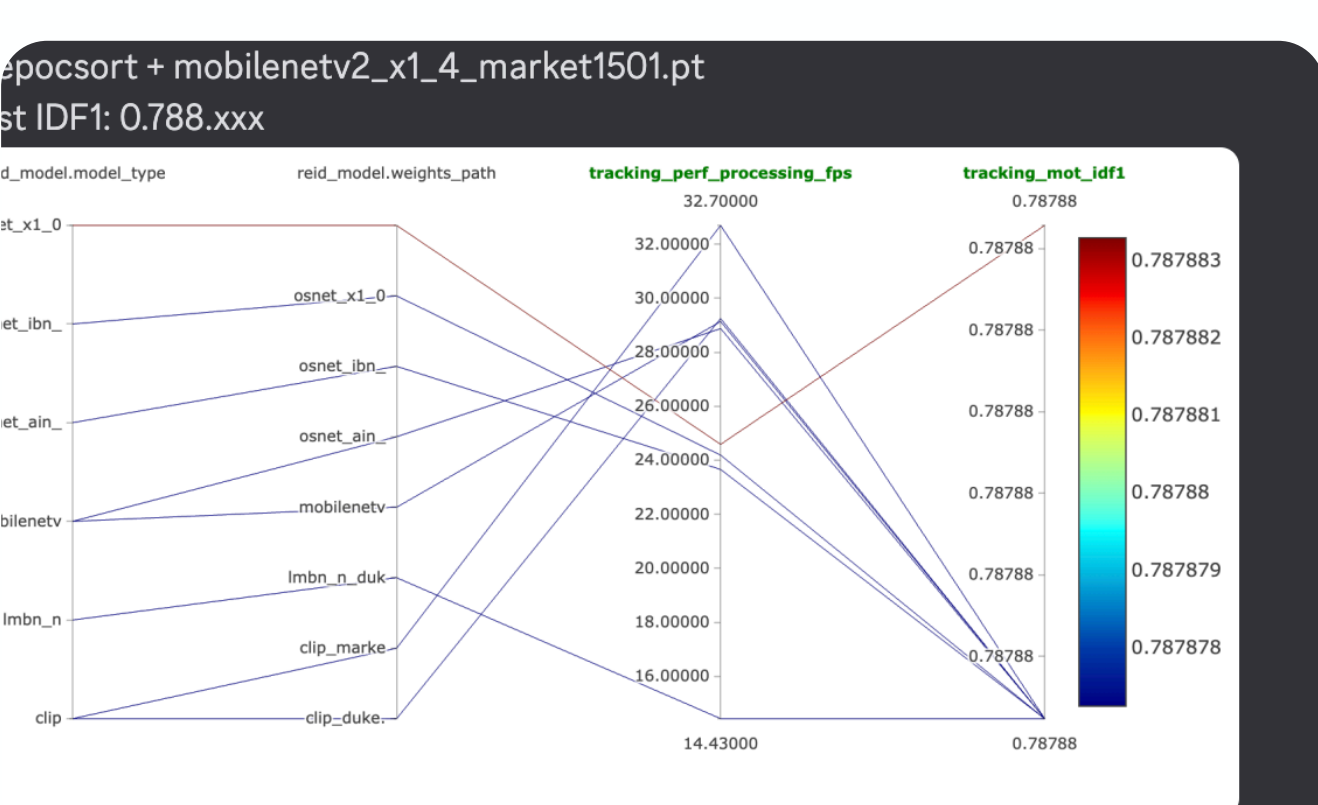
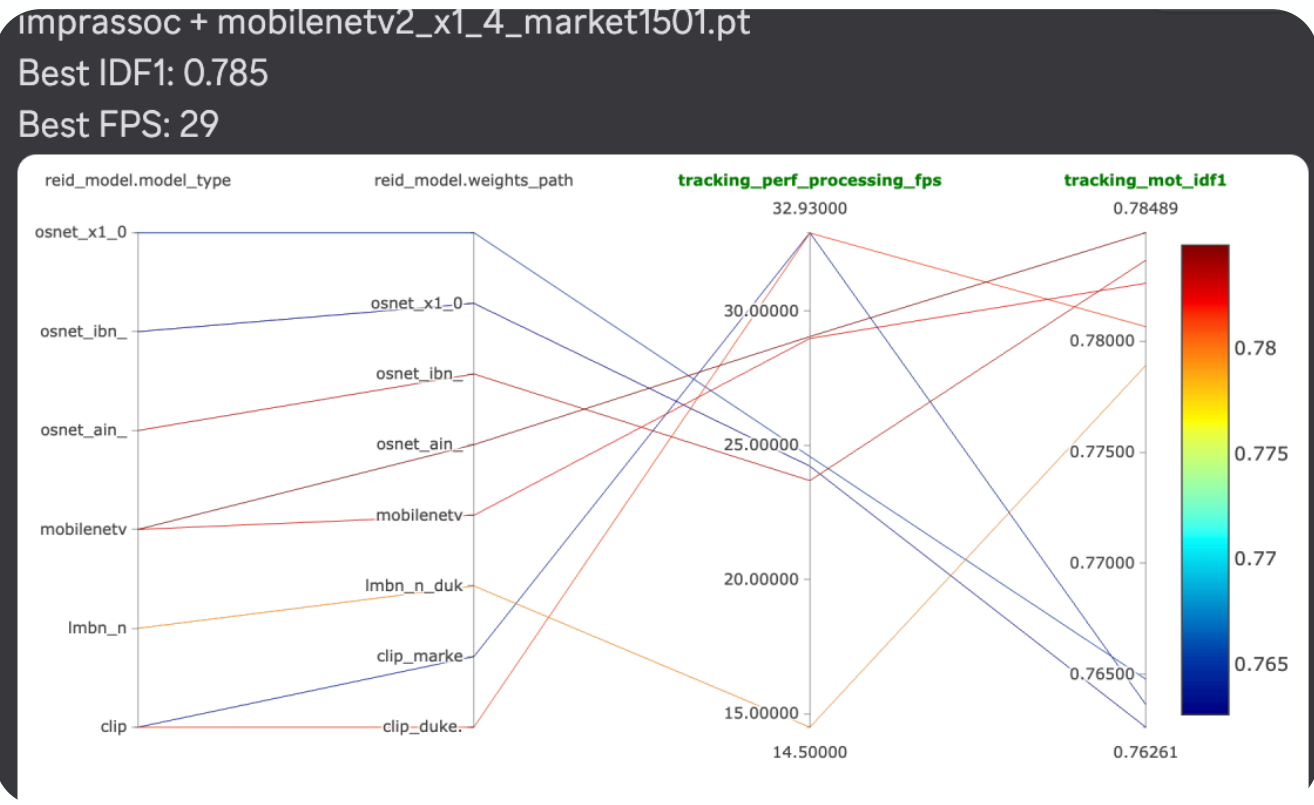
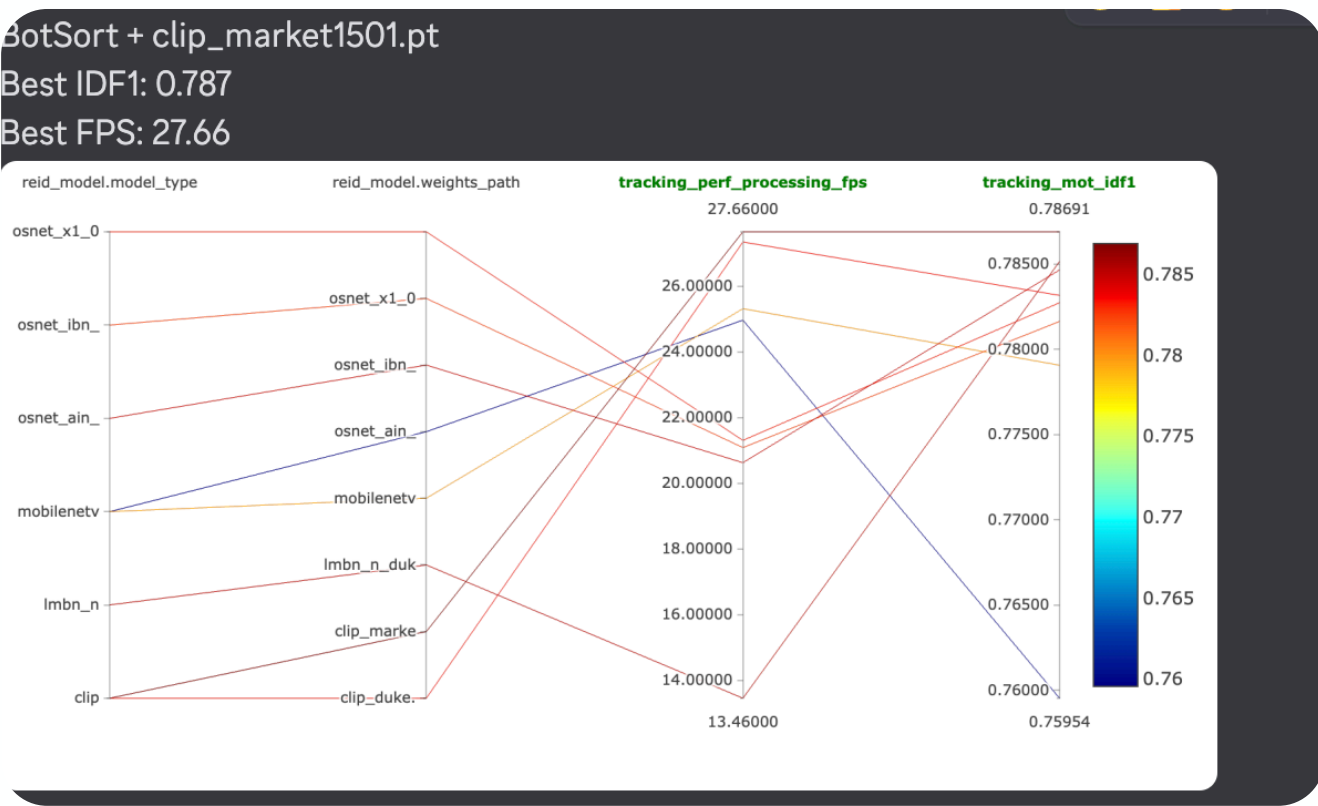
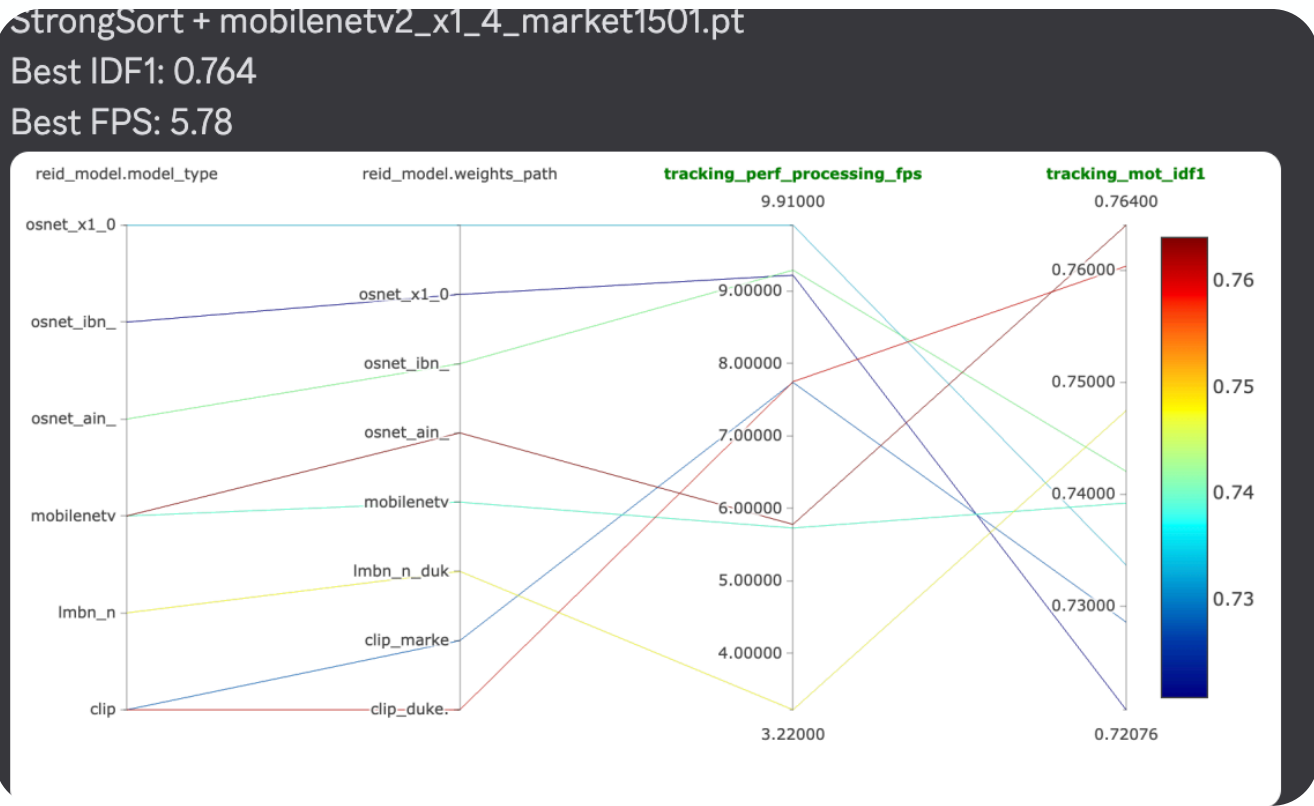
Results (BotSort)



Results (StrongSort)



Results



Conclusion

- BotSort + Clip Market

Tracker	Status	HOTA↑	MOTA↑	IDF1↑	FPS
boosttrack	✓	69.253	75.914	83.206	25
botsort	✓	68.885	78.222	81.344	46
strongsort	✓	68.05	76.185	80.763	17
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ByteTrack vs BotSort

Feature	ByteTrack	BoT-SORT
Low-Score Detections	Core Strategy: Actively uses low-score detections in a second matching stage to recover occluded/difficult objects.	Primarily relies on high-score detections for initial association. Low-score detections are not a central part of its core matching strategy.
Association Logic	Motion-First: Primarily motion-based (IoU matching). Appearance (ReID) is often secondary or optional.	Motion + Appearance: Integrates strong appearance-based ReID with motion cues (from Kalman Filter & IoU) for robust identity maintenance.
Kalman Filter	Standard Kalman Filter for state prediction (position, velocity).	Advanced Kalman Filter for more precise motion and object size prediction.
Camera Motion Comp.	Relies on tracker robustness; no dedicated CMC module.	Explicitly includes Camera Motion Compensation (CMC) for stability with moving cameras.

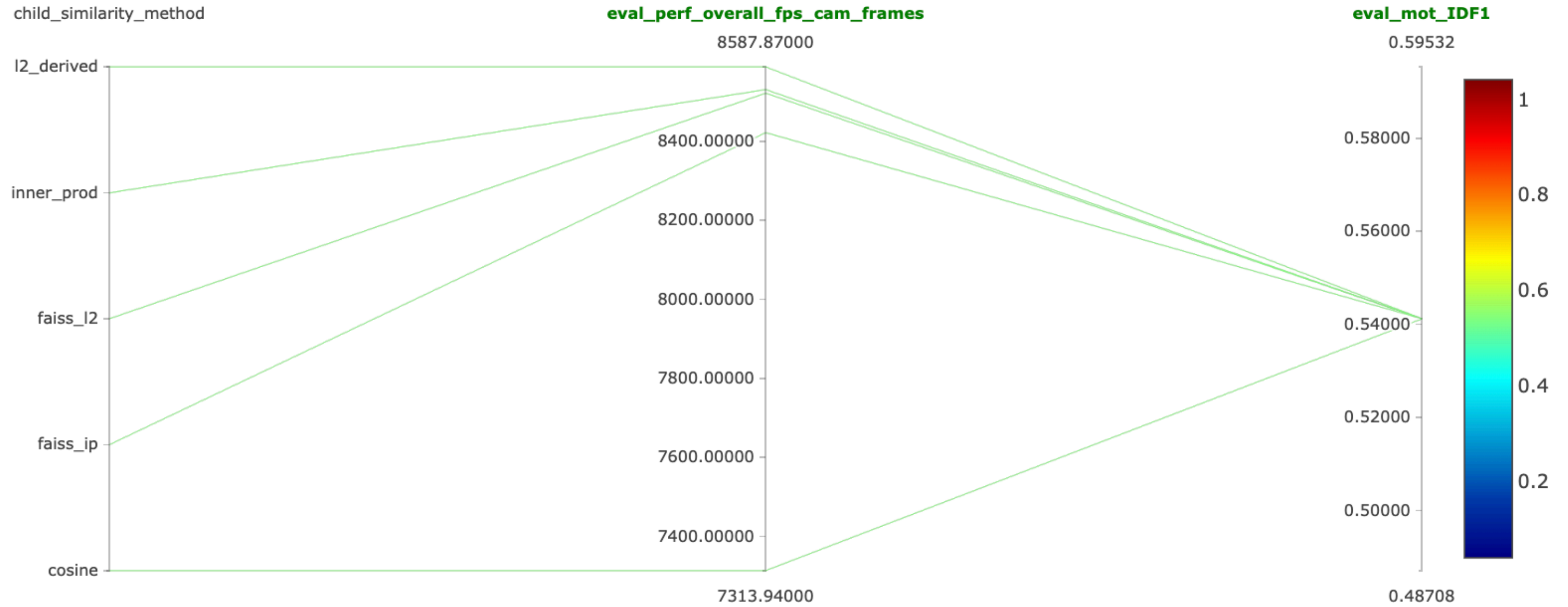
Position, velocity

Position, velocity, and object dimensions (size)

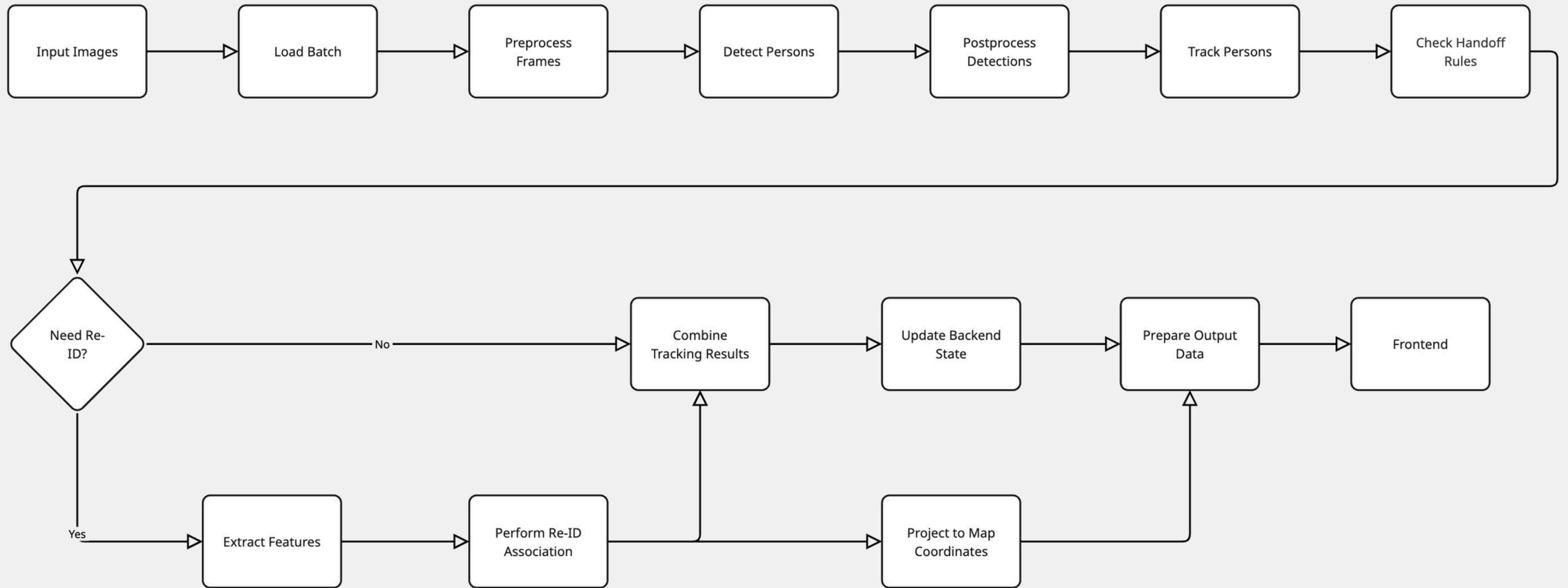
Similarity Search

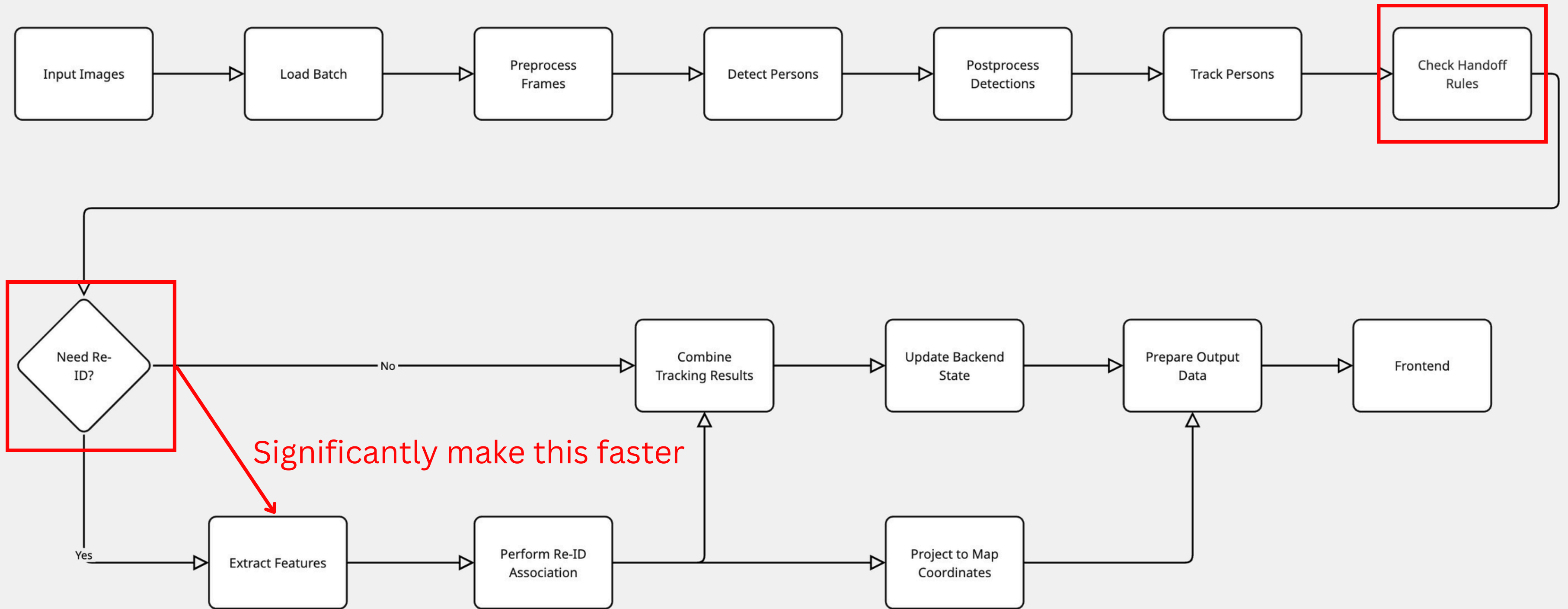
Experiments

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Workflow refinement





Interesting topic from the final exam

AI-Enabled

ML Canvas



Model Fairness Analysis

Task 1.3: Model Fairness Analysis

Identifying Potential Biases in Training Data

- **Demographic Imbalance:**

- Geographic/Ethnic (KAIST, Korea → predominantly East Asian).
- Age/Gender (Campus/factory distribution may not be general).
- Clothing Style (Campus attire).

- **Environmental & Situational Bias:**

- Lighting Conditions (Underrepresentation of very low light/harsh shadows).
- Camera Viewpoint/Quality (Distant, oblique, low-res views).
- Occlusion Levels.
- Behavioral Patterns.

Grad-CAM

Task 1.5: Model Explainability

Grad-CAM Output – Factory Scene

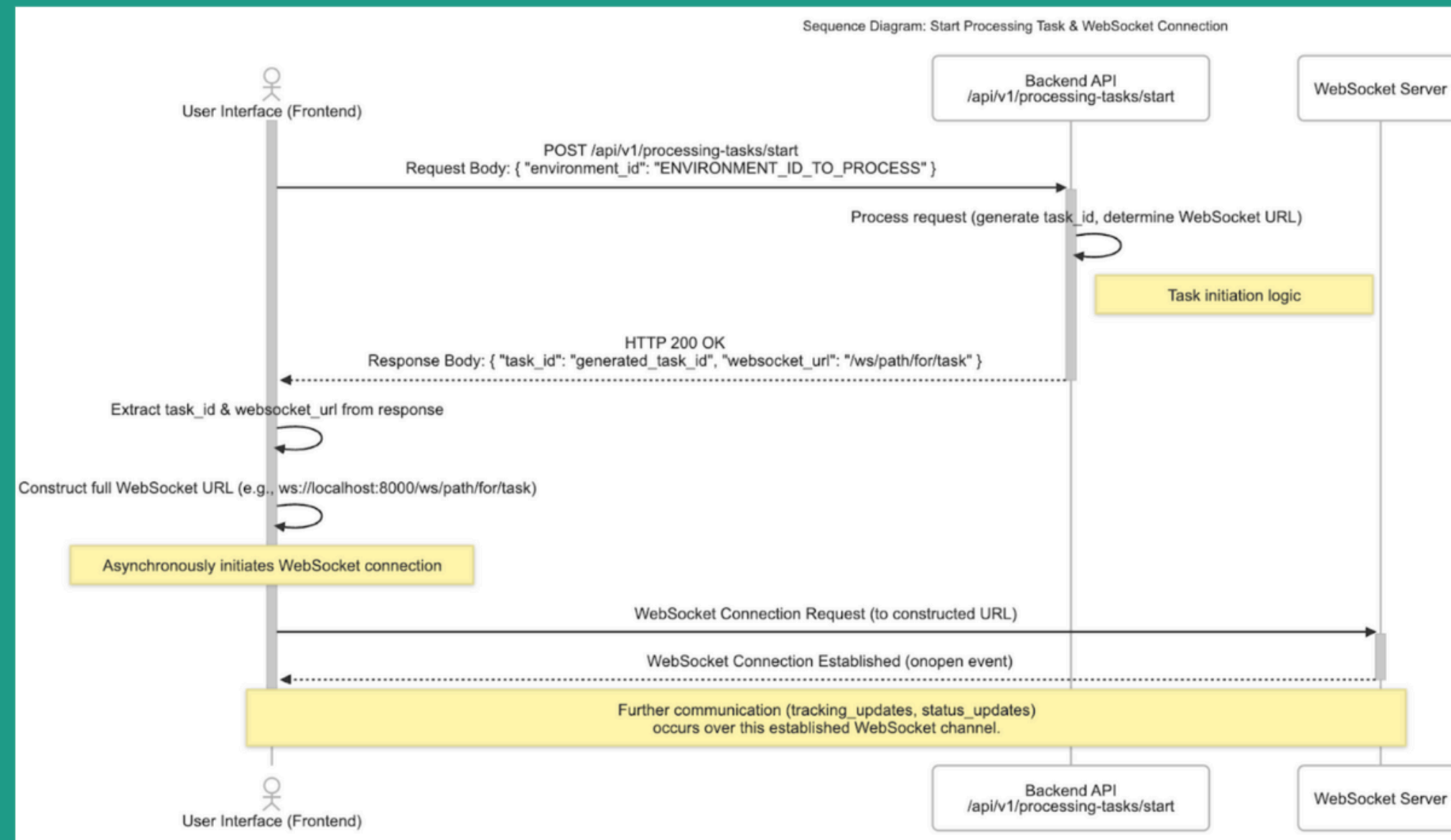
Explanation Heatmap (Explaining Class: 1, Score: 1.000 Detection)



Sequence Diagram

Task 2.2: UI – Model Interface Design

Sequence Diagram Showing UI-Backend Interaction



Model Deployment

```
{
  "type": "tracking_update",
  "payload": {
    "frame_index": 123, // 0-indexed global frame counter for
    the task
    "scene_id": "string", // Corresponds to the
    environment_id (e.g., "factory")
    "timestamp_processed_utc": "string (isoformat datetime)",
    "cameras": {
      "camera_id_string_1": { // e.g., "c09"
        "image_source": "string", // e.g., "000123.jpg"
        (pseudo-filename)
        "tracks": [
          {
            "track_id": "number (int)", // Intra-camera track
            ID
            "global_id": "string (uuid or null)", //
            System-wide unique person ID
            "bbox_xyxy": "[x1, y1, x2, y2]", // list of
            floats
            "confidence": "number (float, optional)",
            "class_id": "number (int, optional)", // e.g., 1
            for person
            "map_coords": "[X, Y]" // list of floats or null
          }
          // ... more tracked persons in this camera
        ],
      },
      "camera_id_string_2": {
        /* ... */
      }
    }
  }
}
```

**Task 1.7: Model
Deployment as a
Service
(WebSocket
Endpoint: Real-time
Tracking Updates)**

Frontend / Backend communication

Demo

Containers / spoton_backend_service

<



fd67b488e8fd



 [spoton_backend_service](#)

STATUS

Running (8 seconds ago)









8000:8000



Logs

Inspect

Bind mounts

Exec

Files

Stats

```

INFO - [Task eec83943-a365-4dcb-bced-d124a9d4d6d6][Env factory] MFProc: Processing batch
for global frame count 2.
2025-05-14 09:25:45 2025-05-14 02:25:45,910 - app.services.multi_camera_frame_processor -
INFO - [Task eec83943-a365-4dcb-bced-d124a9d4d6d6][c12] HANDOFF TRIGGER: Track ('c12', 1
) matched rule 'left' -> Cam 'c13' Area 'upper left'. Overlap: 1.00
2025-05-14 09:25:47 2025-05-14 02:25:47,158 - app.services.multi_camera_frame_processor -
INFO - [Task eec83943-a365-4dcb-bced-d124a9d4d6d6][c13] HANDOFF TRIGGER: Track ('c13', 2
) matched rule 'right' -> Cam 'c09' Area 'down'. Overlap: 1.00
2025-05-14 09:25:47 2025-05-14 02:25:47,159 - app.services.multi_camera_frame_processor -
INFO - [Task eec83943-a365-4dcb-bced-d124a9d4d6d6][c13] HANDOFF TRIGGER: Track ('c13', 1
) matched rule 'left' -> Cam 'c12' Area 'upper left'. Overlap: 1.00
2025-05-14 09:25:48 2025-05-14 02:25:48,988 - app.services.multi_camera_frame_processor -
INFO - [Task eec83943-a365-4dcb-bced-d124a9d4d6d6] MFProc: Batch for frame count 2 finis
hed in 43.550s. Active tracks: 24. Features for ReID: 24. Handoff Triggers: 3.
2025-05-14 09:25:49 2025-05-14 02:25:49,000 - app.services.pipeline_orchestrator - INFO -
Task eec83943-a365-4dcb-bced-d124a9d4d6d6 status: PROCESSING - Progress: 0.00%
2025-05-14 09:25:49 2025-05-14 02:25:49,121 - app.services.pipeline_orchestrator - INFO -

```











```

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    ...
2025-05-14 09:27:58,235 - websocket_client - INFO - Details: Task eec83943-a365-4dcb-bced-d124a9d4d6d6
for environment 'factory' queued.
2025-05-14 09:27:58,330 - websocket_client - INFO - [TASK eec83943-a365-4dcb-bced-d124a9d4d6d6] [STATUS_U
PDATE]
2025-05-14 09:27:58,331 - websocket_client - INFO - Status: PROCESSING
2025-05-14 09:27:58,331 - websocket_client - INFO - Progress: 0.00%
2025-05-14 09:27:58,332 - websocket_client - INFO - Current Step: Processing sub-video 1, frame batch
6 (Global Index: 5)
2025-05-14 09:27:58,332 - websocket_client - INFO - Details: Task eec83943-a365-4dcb-bced-d124a9d4d6d6
for environment 'factory' queued.
2025-05-14 09:28:41,471 - websocket_client - INFO - [TASK eec83943-a365-4dcb-bced-d124a9d4d6d6] [TRACKING
_UPDATE] Frame: 5, Scene: factory, TS: 2025-05-14T02:28:41.465171+00:00
2025-05-14 09:28:41,471 - websocket_client - INFO - Camera: c09, ImgSrc: 000005.jpg, Tracks: 4
2025-05-14 09:28:41,471 - websocket_client - INFO - -> Person 1: TrackID: 1, GlobalID: 56dab09a-534b
-4e1b-a349-c63326522ac8, BBox: [1410.9, 258.6, 1527.5, 444.9], Conf: 0.9807141423225403, ClassID: 1, Map
Coords: [1469.2, 444.9]
2025-05-14 09:28:41,471 - websocket_client - INFO - -> Person 2: TrackID: 2, GlobalID: 0f26a408-7294
-4b9f-bd02-084904330ea0, BBox: [1035.4, 234.6, 1138.2, 418.4], Conf: 0.9778677225112915, ClassID: 1, Map
Coords: [1086.8, 418.4]
2025-05-14 09:28:41,471 - websocket_client - INFO - -> Person 3: TrackID: 3, GlobalID: 14ea9e11-aae8
-4e66-8248-53477b319b0d, BBox: [1805.0, 274.1, 1902.7, 462.1], Conf: 0.9641561508178711, ClassID: 1, Map
Coords: [1853.8, 462.1]
2025-05-14 09:28:41,471 - websocket_client - INFO - -> Person 4: TrackID: 4, GlobalID: 615feb45-c1d5
-41b2-8e51-e8b31035cc4b, BBox: [1358.4, 188.6, 1419.0, 376.7], Conf: 0.9048377871513367, ClassID: 1, Map
Coords: [1388.7, 376.7]
2025-05-14 09:28:41,471 - websocket_client - INFO - Camera: c16, ImgSrc: 000005.jpg, Tracks: 12
2025-05-14 09:28:41,472 - websocket_client - INFO - -> Person 1: TrackID: 1, GlobalID: bd9d0a17-68e5
-4d6f-9686-f963dbbcd469, BBox: [101.8, 300.4, 283.3, 636.3], Conf: 0.9993368983268738, ClassID: 1, MapCo
ords: [192.5, 636.3]
2025-05-14 09:28:41,472 - websocket_client - INFO - -> Person 2: TrackID: 2, GlobalID: 8c43343a-3cd9
-40e4-81fe-138a4a796037, BBox: [391.0, 636.6, 676.5, 997.7], Conf: 0.9966897964477539, ClassID: 1, MapCo
ords: [533.8, 997.7]
2025-05-14 09:28:41,472 - websocket_client - INFO - -> Person 3: TrackID: 3, GlobalID: b7962624-850b
-4252-4252-4252-4252, BBox: [425.2, 636.6, 705.2, 997.7], Conf: 0.9966897964477539, ClassID: 1, MapCo
ords: [577.7, 997.7]

```

< Back

Campus (Task: e189e036...)

Num cameras: 4

● Active: 4

● Inactive: 0

Map: floor 1

Play All

Stop All

View all

Camera 1

Camera 2

Camera 3

Camera 4

ID: 89f515f-531a-44dc-bap-3
ID: 635ca5ae-0b91-
ID: 1ca23466 ID: 44592139-dc28-4e91-9571-d7aea

ID: 30efcb24-e9a7-4eb0-a32a-741ecf7f13973

ID: a93b2f0e-b4c1-4cb3-8da1-8f26e6ffea11
ID: 1ca23466 ID: 44592139-dc28-4e91-9571-d7aea

ID: 6c2e120d-b9eb-4d92-8439-ae15fa740f7e

Camera 1

Camera 2

Camera 3

Camera 4

Detections per camera

Camera 1

145

Camera 2

117

Camera 3

82

Camera 4

29

Last caught ● Active

Camera 3

Person Id: 1

Tracking start: 25 min. ago

Last known: Room1

Cancel Tracking

Internship plan

Plan & Timeline

Internship (6 months)

- **First 3 months → focus on internship.**
- **Meeting during last 3 months, once a month**
- **Working on project**

Second Semester

- **Continue working on project**

